

CODE AND REPRODUCTION GUIDE

TECHNICAL APPENDIX FOR HONORS THESIS: GEOPOLITICAL RISK AND GLOBAL COMMODITIES: AN EMPIRICAL STUDY.

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This appendix maps the MATLAB replication scripts to the figures and tables in the thesis. The empirical analysis combines a two-step SVARX specification, a LASSO-based robustness exercise, and single-equation autoregressive distributed lag (ADL) impulse responses. The nonlinear shock-transmission implementation draws on the MATLAB code and methodology of Forni et al. (2024), adapted here to geopolitical risk shocks. The ADL analysis follows the model-selection approach of Baek and Lee (2022).

The website and the Github Repository of this project could be accessed at [here](#) and [here](#).

0.1 Folder setup

Set the MATLAB current directory to the replication folder. Keep the Data and Functions subfolders, input .mat files, and country data scripts at the locations specified in the MATLAB code. For example:

```
cd('/path/to/replication-folder')
addpath('Data')
addpath('Functions')
Fig123
```

Run each top-level script by entering its filename without the .m suffix. Figures appear in MATLAB unless the script explicitly exports them; tabular output may be printed in the Command Window. Each script can be run separately when its inputs and helper functions are available.

0.2 Main-text outputs

Table 1: MATLAB scripts for the main-text figures and tables

Script	Output	Description
Fig123.m	Figures 1–3	Country-specific GPR standard deviations and the full-sample and recent-period correlation heatmaps.
Plot_F67.m	Figures 6–7	Historical geopolitical act and threat shocks, together with their squared shocks.
Main.m	Figures 8–10	World GPR SVARX responses: linear, quadratic, and overall components.
Main_Lasso.m	Figures 11–13	Monthly aggregate, act, and threat comparisons. LASSO responses are shown as blue dashed lines with light-blue confidence bands.
VAR_Multicountries.m	Figures 14–16	Multi-country group impulse response comparisons.
ShockExam.m	Tables 2–3	Diagnostics for Cholesky-identified and LASSO-extracted GPR shocks.
ADL_Multicountry.m	Figures 17–23	Global ADL responses (Figures 17–19) and cross-country responses of the policy rate, one-year yield, real activity, and gold production (Figures 20–23).

0.3 Appendix outputs

Table 2: MATLAB scripts for the supplementary results

Script	Output	Description
ShockExam.m	Tables 6–7	Full Newey–West predictability regressions for Cholesky-identified and LASSO-extracted shocks (Appendix A.1).
Main_Lasso.m	Figures 24–29	Supplementary linear and quadratic responses for aggregate GPR, acts, and threats (Appendix B.1).
VAR_Multicountries.m	Figures 30–31	Additional country-group responses for the U.S. and Pacific allies, and for China, Korea, and Japan (Appendix B.2).

Figures 4–5 (the Caldara–Iacoviello replication) are not assigned to these seven scripts in the supplied script map. The replication code to the aforementioned could be downloaded straightly from [here](#). Appendix C.1 contains a mathematical proof of the ADL transfer function rather than a separate generated figure.

0.4 Methodological references

In Section 3.2 of the thesis, a preliminary VAR identifies a GPR innovation. That innovation and its square enter the SVARX response analysis as exogenous variables, allowing the linear and quadratic contributions to be displayed separately and combined. The implementation is informed by Forni, Gambetti, Maffei-Faccioli, and Sala (2024), whose original application analyzes financial shocks. The data and empirical estimates in this thesis concern geopolitical risk shocks.

In Section 4, the ADL model estimates each response equation separately. Following Baek and Lee (2022), the distributed-lag length for the shock is tied to the impulse-response horizon, while the autoregressive lag order is selected using BIC. The ADL figures provide a robustness comparison with the system-based SVARX estimates.

Code and method references. Forni, M., Gambetti, L., Maffei-Faccioli, N., and Sala, L. (2024), “Nonlinear Transmission of Financial Shocks: Some New Evidence,” *Journal of Money, Credit and Banking*, 56(1), 5–33, doi:10.1111/jmcb.13099. Baek, C., and Lee, B. (2022), “A Guide to Autoregressive Distributed Lag Models for Impulse Response Estimations,” *Oxford Bulletin of Economics and Statistics*, 84(5), 1101–1122, doi:10.1111/obes.12494.